Set 21: Energy Consumption Analysis

Data Visualization using R

Aim

To analyse energy consumption data in R using a histogram, a density plot, a bubble scatter plot, a bar chart and a dashboard with filters.

Procedure

22. Open RStudio and load the packages ggplot2 and gridExtra (install them once with install.packages(c("ggplot2", "gridExtra"))).
23. Save the data as Energy_Consumption_Data.csv and import it with read.csv().
24. Question 1: plot the histogram with geom_histogram() and the density curve with geom_density().
25. Question 2: draw the scatter plot of temperature against units with bubble size = Peak_Hours and alpha for transparency.
26. Question 3: find the average Renewable_Usage of each sector with aggregate() and draw the bar chart.
27. Question 4: draw the monthly line chart and the Sector vs Region heatmap, join them into a dashboard, and apply the Month and Sector filters with subset().
28. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)
options(width = 95)

# Create the CSV file and import it
data.frame(
  Sector = c("Residential","Commercial","Industrial","Residential","Commercial","Industrial"),
  Region = c("North","South","West","East","North","South"),
  Month = c("Jan","Jan","Feb","Feb","Mar","Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
) -> d
write.csv(d, "Energy_Consumption_Data.csv", row.names = FALSE)
energy <- read.csv("Energy_Consumption_Data.csv")
energy$Month <- factor(energy$Month, levels = c("Jan","Feb","Mar"))
print(energy)

# 1. Histogram and density plot
print(aggregate(Units_Consumed ~ Sector, energy, mean))
p1 <- ggplot(energy, aes(Units_Consumed)) +
  geom_histogram(aes(y = after_stat(density)), binwidth = 150, fill = "skyblue", colour = "white") +
  geom_density(colour = "red") +
  labs(title = "Histogram and Density of Units Consumed", x = "Units Consumed (kWh)", y = "Density")

# 2. Scatter plot with bubble size and transparency
cat("Correlation:", cor(energy$Temperature, energy$Units_Consumed), "\n")
p2 <- ggplot(energy, aes(Temperature, Units_Consumed, size = Peak_Hours))
+ geom_point(alpha = 0.5, colour = "blue") +
  labs(title = "Temperature vs Units Consumed", x = "Temperature (C)", y = "Units Consumed (kWh)")

# 3. Average renewable usage by sector
avg <- aggregate(Renewable_Usage ~ Sector, energy, mean)
print(avg)
```

```
p3 <- ggplot(avg, aes(Sector, Renewable_Usage, fill = Sector)) + geom_col(show.legend = FALSE) +
  geom_text(aes(label = paste0(Renewable_Usage, "%")), vjust = -0.5) +
  labs(title = "Average Renewable Usage by Sector", y = "Renewable Usage (%)")

# 4. Dashboard (line chart + heatmap); filters are applied with subset()
dash <- function(d, title) {
  line <- ggplot(aggregate(Units_Consumed ~ Month, d, sum), aes(Month, Units_Consumed, group = 1)) +
    geom_line() + geom_point(size = 3) + labs(title = "Monthly Energy Usage", y = "Units (kWh)")
  heat <- ggplot(d, aes(Region, Sector, fill = Units_Consumed)) + geom_tile() +
    geom_text(aes(label = Units_Consumed)) + scale_fill_gradient(low = "yellow", high = "red") +
    labs(title = "Heatmap: Sector vs Region", fill = "Units")
  arrangeGrob(line, heat, ncol = 2, top = title)
}
d1 <- dash(energy, "Energy Dashboard (No Filter)")
d2 <- dash(subset(energy, Month %in% c("Jan","Feb") & Sector != "Industrial"),
           "Energy Dashboard (Filter: Jan-Feb, Residential and Commercial)")

ggsave("set21_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set21_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set21_q3.png", p3, width = 6, height = 4, bg = "white")
ggsave("set21_q4a.png", d1, width = 12, height = 5, bg = "white")
ggsave("set21_q4b.png", d2, width = 12, height = 5, bg = "white")
```

Output

```
      Sector Region Month Temperature Units_Consumed Cost Renewable_Usage Peak_Hours
1 Residential  North   Jan          15           320 2100              22          4
2  Commercial  South   Jan          24           540 3600              18          6
3  Industrial   West   Feb          20           880 5900              12          8
4 Residential   East   Feb          18           350 2300              25          5
5  Commercial  North   Mar          28           610 4100              20          7
6  Industrial  South   Mar          30           920 6200              15          9
      Sector Units_Consumed
1  Commercial            575
2  Industrial            900
3 Residential            335
Correlation: 0.654655
      Sector Renewable_Usage
1  Commercial           19.0
2  Industrial           13.5
3 Residential           23.5
```

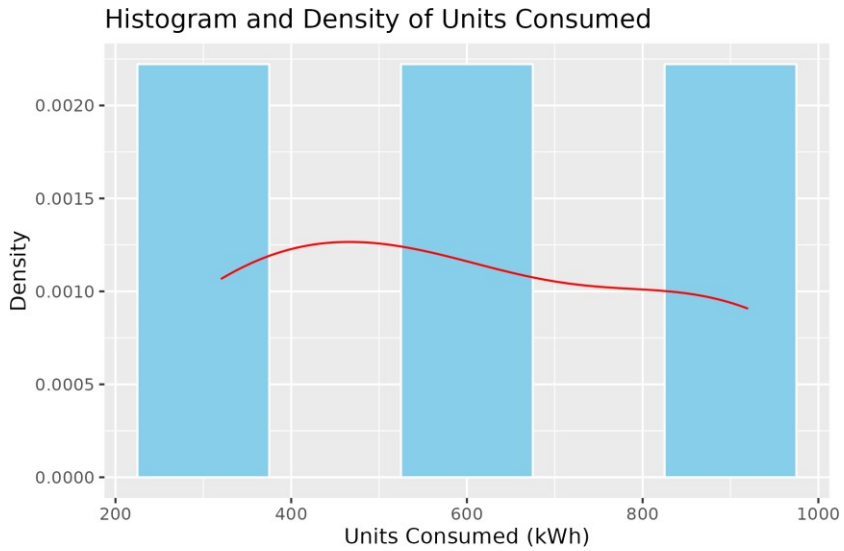


Figure 21.1: Q1 - Histogram and density plot

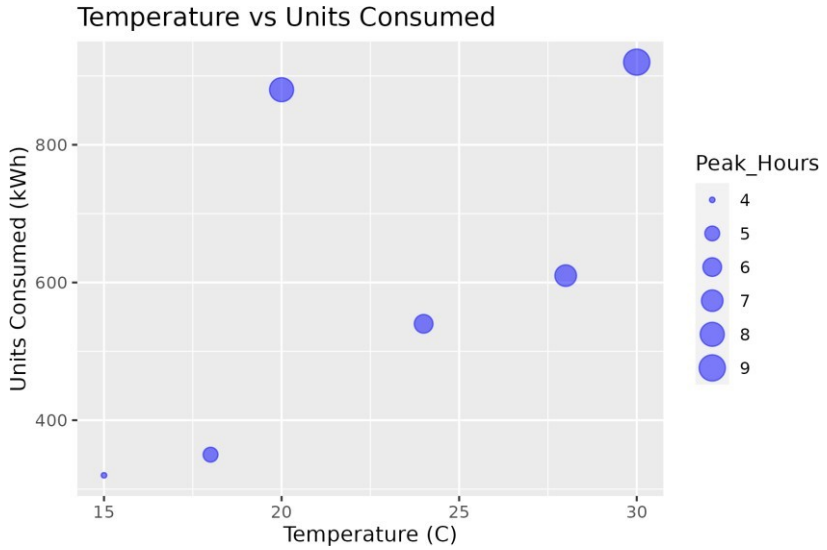


Figure 21.2: Q2 - Temperature vs units (bubble = peak hours)

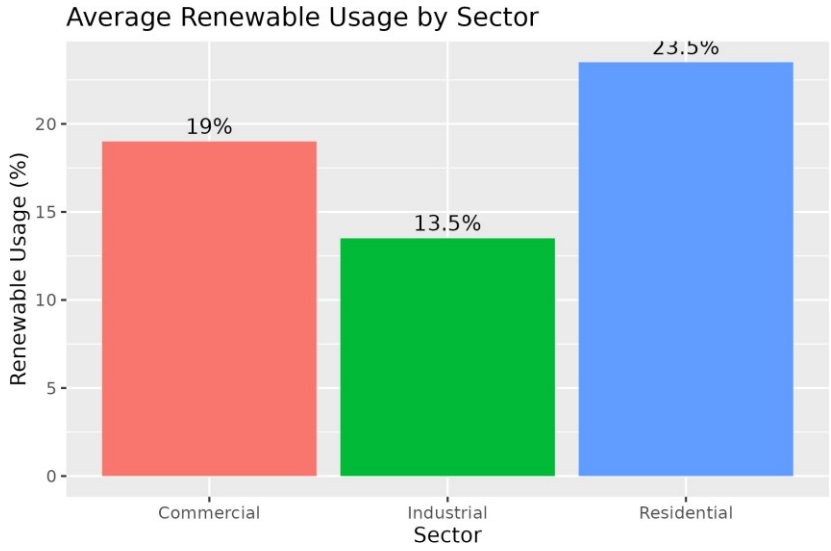


Figure 21.3: Q3 - Average renewable usage by sector

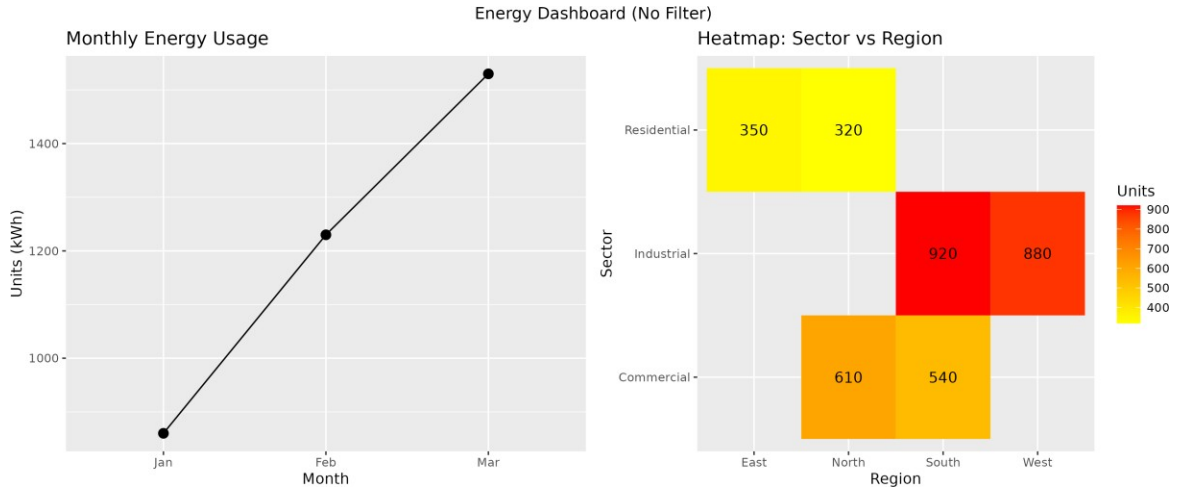


Figure 21.4: Q4 - Dashboard (no filter)

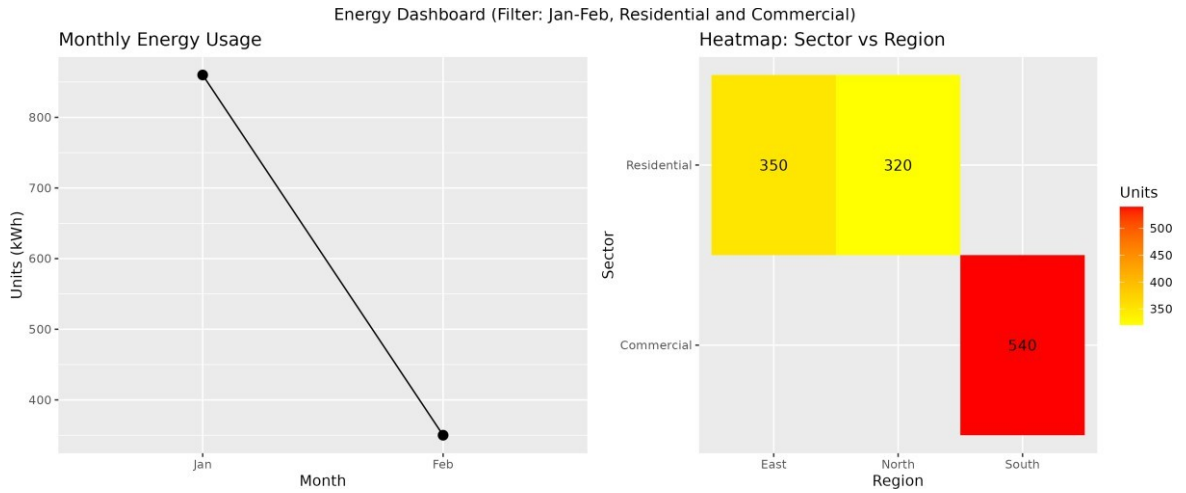


Figure 21.5: Q4 - Dashboard (filter: Jan-Feb, Residential and Commercial)

Result

- Industrial uses the most energy (average 900 kWh) and Residential the least (335 kWh).
- Temperature and units have a positive correlation of 0.65.
- Residential has the highest renewable usage (23.5%) and Industrial the lowest (13.5%), so Industrial has the most scope to improve.
- Energy use rises from January to March, and the heatmap shows the highest use in Industrial.